The residually weakly primitive geometries of the Hall–Janko group

Human Compression project

Abstract

We recover the fixed principal census of Leemans for the Hall–Janko group: eight classes in rank two, nineteen in rank three and three in rank four, with none in any higher rank. The common recognition and augmentation theorems account for arbitrary chamber stabilizers, including the nontrivial one occurring in rank four. Complete exact programs enumerate, fuse and independently verify the returned geometries, conditional on the stated subgroup catalogue and finite-group algorithms.

The source is D. Leemans, *The residually weakly primitive geometries of* J_2, Note di Matematica **21** (2002), 77–81. We use its five-page journal article. The fixed target is the complete census under its stated axioms and equivalence, not every description of the classified objects.

A target geometry is firm, flag transitive and residually connected. It is residually weakly primitive (RWPRI): in each residue some type action of its flag stabilizer is primitive. Its panel actions are two-transitive, the property denoted $(2T)_1$, and its rank-two residues are partial linear spaces or generalized digons, denoted $(\mathrm{IP})_2$. The latter is a geometric condition, distinct from subgroup intersection identities. The equivalence permits all automorphisms of J_2 and all permutations of the types. It concerns the specified group action; no identification of the full automorphism group of every incidence structure is inferred from it.

The complete theorem UD.PanelRecognition translates these axioms into the actual panel subgroups and their coset actions. Theorem UD.MaximalAugmentation then reduces classification to stabilizer actions on extension fibers and proves termination by strict subgroup chains. These common proofs are paid in full in this topic. The following application supplies their finite data and verifies every local condition.

1 The complete census

Apply UD.MaximalAugmentation to $G = J_2$. The retained TomLib catalogue has 146 subgroup classes and 568 maximal-inclusion classes, of which 417 are locally two-transitive. These are finite-group classification data, not a geometry list. The complete search tests 4,072 maximal-augmentation orbit representatives, distributed by ranks two through six as $644, 1226, 1931, 271, 0$. Every one of its 146 Borel anchors reaches an empty terminal level.

Rank	Inner conjugacy and type permutations	Aut(J2) and type permutations
2	8	8
3	24	19
4	5	3
At least 5	0	0

The final fusion uses the full automorphism group and exact successive subgroup transporters, not just equality of subgroup-order signatures. All 30 final representatives pass a separate check through

their maximal parabolics: generation by panels, reconstruction of every panel by intersections, firmness, typed Helly, all residue-generation conditions, all RWPRI deletions, two-transitive panels and every rank-two partial-linearity/digon test. The three rank-four Borel orders are 1,1,8; their maximal-parabolic order multisets are respectively [12,150,60,60], [150,150,60,60] and [384,384,6048,6048].

Corollary D3.J2. Under the stated complete subgroup catalogue and exact finite-group operations, these are all firm flag-transitive residually connected J_2 geometries satisfying RWPRI, $(2T)_1$ and $(\mathrm{IP})_2$: eight in rank two, nineteen in rank three and three in rank four, with none in any higher rank.

Proof. UD.MaximalAugmentation proves exhaustion, including proper acting subgroups during intermediate stages. The complete finite output and separate final-witness tests give existence and exact full-automorphism/type equivalence. An empty terminal level at every Borel anchor excludes every further rank. This is not an inference from the absence of a rank-five object alone. □

2 Inputs, equivalence and complete finite evidence

The search uses the complete subgroup-class and maximal-inclusion data for J_2 supplied by GAP's TomLib. It does not use a list of known geometries. For every subgroup B, conjugate inclusions are reconstructed by UD.Transport. The action of $N_{J_2}(B)$ on the actual panel subgroups may have a kernel; replacing it by its permutation image preserves the orbits because every image element has an acting lift. All incidence and subgroup tests are still performed on the original subgroups. No faithfulness of this auxiliary action is assumed.

For a final geometry, the action of J_2 itself is faithful. Its kernel is a normal subgroup contained in every proper type stabilizer. Simplicity therefore makes this kernel trivial. This fact is separate from the possible kernel of a normalizer action used to accelerate the enumeration.

The full programs are the search, the fusion under $\mathrm{Aut}(J_2)$ and type permutations, the final verification, and the shared geometry tests. Each of the 146 Borel anchors must have a completed checkpoint whose next level is empty. Intermediate geometries for proper acting subgroups are retained as required by the induction; stopping after finding the desired rank-four objects would not prove completeness. The final fusion uses transporters between actual subgroups, not a comparison of their orders alone.

The replay driver checks pinned input bytes, creates a fresh working tree, checks every anchor and empty terminal level, and checks all thirty final witnesses. The historical certificate-builder script is not invoked. Exact GAP algorithms and completeness of the named subgroup data remain retained inputs. The full programs, their outputs and those assumptions form the computational proof; the table of totals is only its summary.

3 The exact source boundary

The source's main article, on printed pages 79–80, also displays rank-four diagrams, local annotations, full geometric automorphism and correlation groups, and a doubling relation between examples. These assertions are not all recovered here. They are printed-body omissions, not merely missing annotations in an external supplement. They remain excluded from the fixed principal census contract. No new historical classification or additional source credit is claimed.

[UD] *Stabilizers, extensions and the enumeration of finite geometries*, common paper in this topic. The arbitrary-chamber-stabilizer recognition and complete augmentation arguments reproduce the previously reviewed proof bank, with the exact finite data retained. No M_{22} result is part of this three-application edition.